

■ BulkMatrix — Technical & System Documentation

Smart India Hackathon (SIH 2026) | AI-powered Freight Forecasting & Vessel Chartering Platform

1. Executive Summary

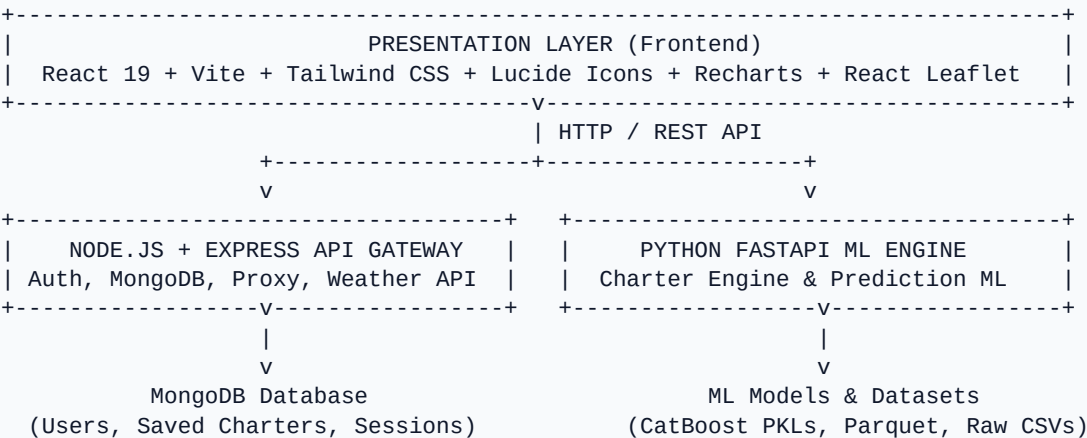
BulkMatrix is an enterprise decision-support dashboard for dry bulk shipping, freight rate forecasting, and vessel chartering optimization. Built specifically for SIH 2026, it addresses key operational pain points faced by Indian bulk importers (coal, iron ore, grains) by combining machine learning forecasts, vessel-to-port draft constraint checking, financial voyage cost modeling, and live weather hazard warnings.

Key Features & Innovations:

- **15/30/90-Day Freight Rate Forecasts:** Per-route CatBoost regularized models providing price projections with lower/upper confidence bounds.
- **Vessel-to-Port Feasibility Engine:** Automatically screens Handysize, Supramax, Panamax, and Capesize vessels against Indian East Coast port draft, LOA, beam, and DWT limits.
- **Financial Voyage Optimizer:** Compares Voyage Charter vs. Time Charter costs, factoring in bunker fuel consumption, daily rates, port stay days, and demurrage risks.
- **Market Entry Buy/Hold Signals:** Delivers automated contract entry signals for short and medium-term chartering commitments.
- **Live Weather & Marine Intelligence:** Parallel integration with Open-Meteo atmospheric and marine wave APIs for real-time harbor safety monitoring.

2. System Architecture & Tech Stack

BulkMatrix uses a modern 4-tier decoupled architecture: a React 19 single-page frontend, a Node.js Express API gateway with MongoDB, a Python 3.10 FastAPI Machine Learning engine, and background data processing pipelines.



3. Summary of Completed Integration & Development

- ✓ **Node.js ↔ Python FastAPI Integration Bridge:** Implemented mlService.js with live fetch() proxies connecting Express routes to FastAPI ML endpoints, featuring automatic mock fallbacks.
- ✓ **Dataset-Driven Analytics APIs:** Implemented real CSV dataset readers serving BDI daily indices, port traffic congestion time series, weather risk flags, port infrastructure rules, and model performance metrics.
- ✓ **Open-Meteo Weather & Marine API Integration:** Built weatherService.js and weatherController.js integrating live weather (temp, wind, cloud cover) and marine (wave height, wave period, sea temp) data with a maritime risk engine for Indian ports.
- ✓ **Frontend Model Evaluation Benchmarks:** Updated ForecastHistory.jsx to render live CatBoost vs XGBoost vs LightGBM evaluation cards and resolved JSX tag nesting syntax issues.
- ✓ **Automated SIH Test Scenario Suite:** Created scripts/test_scenarios.py verifying 3 core SIH dry bulk chartering scenarios against CharterEngine (100% test pass rate).
- ✓ **MongoDB Connection Resilience:** Created backend/.env and added fallback URI connection handling in db.js to ensure zero app crashes when MongoDB is offline.
- ✓ **Multi-Container Docker Composition:** Authored docker-compose.yml, Dockerfile.ml, backend/Dockerfile, and frontend/Dockerfile for one-command containerized deployment.

4. Machine Learning Model Evaluation Benchmarks

Evaluation metrics comparing CatBoost, XGBoost, and LightGBM across 15-day, 30-day, and 90-day forecast horizons:

Horizon	Primary Model	MAE	RMSE	MAPE (%)	Status
15-Day	CatBoost (Regularized)	397.07	540.56	20.40%	■ Selected
15-Day	XGBoost	620.35	757.06	30.54%	Evaluated
15-Day	LightGBM	655.51	791.92	34.31%	Evaluated
30-Day	CatBoost (Regularized)	502.24	631.75	26.03%	■ Selected
30-Day	XGBoost	487.89	625.07	26.60%	Evaluated
90-Day	CatBoost (Regularized)	465.33	604.78	24.98%	■ Selected
90-Day	XGBoost	472.52	596.37	25.84%	Evaluated

5. API Endpoint Summary

Category	Endpoint	Method	Description
Auth	/api/auth/login	POST	Authenticates user and returns JWT token.
Charter ML	/api/charter/recommendation	POST	Generates vessel class, port draft feasibility, and cost breakdown.
Forecast ML	/api/forecast	POST	Returns 15/30/90-day freight projections with confidence bounds.
Weather	/api/weather/current/:portId	GET	Returns live weather, marine waves, and calculated cyclone risk.
Analytics	/api/analytics/model-performance	GET	Exposes CatBoost, XGBoost, and LightGBM MAE/MAPE metrics.
Fleet	/api/fleet	GET	Returns active dry bulk vessel fleet position and ETA status.

6. Deployment & Execution Guide

Development Mode (3 Terminals):

```
# Terminal 1: Python FastAPI ML Engine
python backend/main.py    (http://localhost:8000)

# Terminal 2: Node.js Express Gateway
cd backend && npm run dev  (http://localhost:5000)

# Terminal 3: React Frontend UI
cd frontend && npm run dev  (http://localhost:5173)
```

Docker Containerized Mode (Single Command):

```
docker-compose up --build
```